

MOHIT VARMA SAGI

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SKILLS & CERTIFICATIONS

Languages: Python (Pandas, NumPy, scikit-learn), SQL, T-SQL, PostgreSQL

ML & Modeling: XGBoost, PyTorch, SHAP, LSTM, CNN, forecasting, feature engineering, statistical modeling

Generative AI & LLMs: LangChain, LangGraph, RAG pipelines, agentic AI, prompt engineering, ChromaDB, FAISS, vector databases, Hugging Face

Data Engineering & MLOps: ETL/ELT, Azure Data Factory, Microsoft Fabric, Databricks, Spark, MLflow, model deployment and monitoring, AWS (Lambda, EC2, S3)

Analytics & BI: Power BI, DAX, Tableau, Excel, KPI dashboards, data quality and governance

Certifications: ML with Python (MIT 6.86x) · Intro to CS (MIT 6.00.1x) · Data Science CDS (Henry Harvin)

EXPERIENCE

Data Scientist, Revenue Cycle Management

Aug 2025 – Present

EyeSouth Partners

Atlanta, GA

- Developed a **RAG** search tool with **LangChain** and **ChromaDB** across 1,000+ payer policy documents, cutting prior authorization research from 3 hours to 5 minutes.
- Engineered **ETL/ELT pipelines** on **Microsoft Fabric** and **Azure Data Factory** for 300+ providers, automating **A/R aging** reporting and uncovering **\$1M+** in recovery opportunities.
- Designed **SQL**-based charge capture validation workflows with stored procedures, reducing recurring **CPT/ICD** billing errors and improving clean claim rates across affiliates.
- Automated extraction of 10,000+ billing codes in **Python**, cutting manual errors by 90% and enabling self-service ad hoc reporting across 3 business units.

Data Analyst, Revenue Cycle Management Intern

May 2025 – Aug 2025

EyeSouth Partners

Atlanta, GA

- Built an **XGBoost** scoring model on 2 years of claims data, applying **feature engineering** and signal selection to surface \$200K+ in recoverable payer underpayments.
- Created a **Power BI** KPI dashboard tracking pending encounters and unbilled claims, cutting stakeholder reporting turnaround by 3 days.

Research Scientist, Applied Data Science

Jan 2025 – May 2025

Southeast Regional Cooperatives

Atlanta, GA

- Trained a hybrid **LSTM** and **CNN** forecasting model on 20,000+ records in **Azure Databricks** with **MLflow** tracking; deployed for seasonal planning and validated through EDA, surfacing 3 yield-driving patterns.
- Implemented a **GPT-4 RAG** pipeline for domain Q&A across 15 research datasets, reducing literature lookup time by 60% and translating findings for technical and non-technical audiences.

Research Analyst, ESG Data Analytics

Aug 2024 – Dec 2024

Truist Financial

Atlanta, GA

- Analyzed 200+ ESG and climate risk metrics across 6 financial institutions in **Python**, exposing regulatory disclosure gaps through **statistical modeling** for board-level leadership.
- Replaced a 40-hour/month manual process with an **AWS Lambda** ETL pipeline and built **Tableau** dashboards for executive strategy reporting.

Operations Analyst Intern, Industrial Analytics

May 2023 – Jul 2023

Hindustan Petroleum Corporation Limited

India

- Deployed an IoT anomaly detection system on **AWS EC2** using Z-score and IQR thresholds, flagging equipment failures across 10 machines before downtime occurred.
- Applied **STL time-series** decomposition on 6 months of sensor data, identifying 3 recurring failure patterns that optimized preventive maintenance schedules.

PROJECTS

Multi-Agent Research Copilot

- Architected a 3-agent **LangGraph** system spanning query decomposition, hybrid **BM25** and vector retrieval, and answer synthesis across 200+ PDFs with cited **RAG** responses.
- Ran validation experiments to stress-test output quality and calibration across diverse query types, shipping the system end-to-end.

Predictive Analytics and Risk Modeling Platform

- Constructed a stacked ensemble with feature engineering on 50,000+ records to reach **89% accuracy**, layering in **SHAP** explainability and **MLflow** experiment tracking with model versioning.
- Delivered a **Power BI** risk-tier dashboard segmenting records into high, medium, and low risk bands so stakeholders could prioritize outreach.

EDUCATION

Georgia State University, J. Mack Robinson College of Business

Dec 2025

M.S. in Data Science and Analytics

National Institute of Technology Calicut

May 2024

B.Tech. in Production Engineering and Management